

Permata Bank Use Case Using Machine Learning / AI

Background:

You are a Data Scientist at PermataBank working on a strategic initiative to design, build, and present a machine learning & AI solution. The management asked you to present one of your impactful initiative in their upcoming monthly BoD Meeting. Your task is to identify and develop a use case that delivers meaningful improvements to the bank—whether by enhancing customer campaigns, optimizing operational processes, or strengthening risk management. You have the freedom to choose any use case and problem statement, as long as your solution demonstrates clear, measurable value and is practically applicable within a banking context. Leveraging AI/LLMs is mandatory and must be included in the solution.

Dataset Information:

For this technical assessment, you will work with an open-source dataset hosted on Kaggle:

Financial Transactions Dataset: Analytics

 <https://www.kaggle.com/datasets/computingvictor/transactions-fraud-datasets/data>

This dataset contains simulated financial transaction records designed for experimentation with machine learning tasks related to banking. Please review the dataset description on Kaggle to understand its structure, schema, and licensing terms. You may apply transformation, feature engineering, or other statistics/ML techniques as needed.

Requirements:

As part of this technical assessment, you are required to prepare a clear and concise **presentation deck** summarizing your work. The main slides should not exceed 8 slides (excluding the cover pages and section dividers). You will present the slides to the Data Science Lead and the related Business Stakeholders. The slides should include (but not limited to):

- Project background and problem statements (including Exploratory Data Analysis)
- ML Model Development and Evaluation
- Proposed solution and projected implementation impact
- AI/LLM used and how it is implemented in the solution

Along with your presentation slides, please also submit the **source code** used to train and implement your model. The code may be provided in any readable format, as long as it can be opened and reviewed without issues.